

4 The hybridoma method is one way to produce large quantities of monoclonal antibody.

(a) Name the type of cell of the immune system (immune cell) that produces antibody.

..... [1]

(b) Figure 4.1 outlines five steps involved in the hybridoma method. The steps are **not** in the correct sequence.

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|----------|--|
| P | screen cells for production of desired monoclonal antibody |
| Q | remove immune cells from spleen and fuse with myeloma cells |
| R | culture cells on a special medium to obtain only hybridoma cells |
| S | allow time for an immune response to occur |
| T | inject small mammal with antigen |

Figure 4.1

Using the letters **P**, **Q**, **R**, **S** and **T** in Figure 4.1, state the correct sequence of steps for the production of monoclonal antibody by the hybridoma method.

..... → → → → [2]

(c) Myeloma cells that have not fused with immune cells live for only a few days in the culture medium. The myeloma cells cannot synthesise purine nucleotides and cannot divide.

(i) Explain why a myeloma cell is **not** able to divide if it cannot produce purine nucleotides.

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..... [2]

(ii) The gene *HPRT* codes for an enzyme involved in the synthesis of purine nucleotides. Myeloma cells produce a non-functioning enzyme because they have a mutation in *HPRT*.

Explain **one** way in which a mutation in gene *HPRT* can lead to the synthesis of a non-functioning enzyme.

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(d) Research has been carried out into the use of a monoclonal antibody, known as VRC01, for the prevention of HIV infection in people who are at high risk of infection.

- VRC01 was infused (injected) into volunteers once every eight weeks for 80 weeks.
- If a volunteer became infected, the strain of HIV was identified.
- For each of the different strains of HIV identified, scientists tested the effectiveness of VRC01 to prevent HIV infection of cells grown in culture.

The results showed that the effectiveness of VRC01 varied depending on the strain of HIV.

VRC01 binds to one part of a viral glycoprotein, gp120, which is located in the HIV envelope. HIV attaches to host cells using gp120.

(i) HIV is the abbreviation used for the enveloped virus that causes HIV/AIDS.

State the full name of the virus that causes HIV/AIDS.

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(ii) Explain why an infusion with monoclonal antibody needs to be repeated at intervals to protect a person from HIV infection.

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(iii) Suggest **and** explain why infusions containing a mixture of monoclonal antibodies could be more successful in preventing HIV infection than using **only** VRC01.

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..... [3]

[Total: 14]